VERIFICATION CORE REPORT

Topic: **Python 3.12 AsyncIO & High-Performance Coroutines** | College: **VIT**
Tech: **Python 3.12.8** | Professor: **sam** | Deck ID: **dcc44a90** | Date: 2026-10-17 11:51:46
Docker Image Digest:
python:3.12-slim@sha256:4a8b21cd839e0123ef456789abcde0123456789abcdef0123456789abcdef01

Metric	Value	Metric	Value
Found vs Executed	5 found / 5 run	Unverified (Not Run)	0
Snippets Verified	2/5	Auto-Healed	2
Needs Review / Fail	0	Non-Deterministic	0
Exercises Proven	3/3	Mutation Score	10000%
Template QA Passed	8/9	Docker Backend Runs	0

1. Code Snippet Sandbox Verification

Slide	Language & Backend	Status	Duration	Captured Output
Slide 3	python 3.12.8 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	264.3556804656982 ms	`Introduction to AsyncIO Executed in Python 3.12`
Slide 4	python 3.12.8 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	268.0953720855713 ms	`Understanding Coroutines Executed in Python 3.12`
Slide 5	python 3.12.8 (isolated_local_subprocess)	AUTO HEALED	2224.7867584228516 ms	`Main function started Main function ended Coroutine 2 started Coroutine 2 ended`
Slide 6	python 3.12.8 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	265.6246393585205 ms	`AsyncIO and I/O Operations Executed in Python 3.12`
Slide 7	python 3.12.8 (isolated_local_subprocess)	AUTO HEALED	61.09142303466797 ms	`Execution verified`

2. Exercise Proof & Mutation Score Matrix

Tier	Exercise Ref	Final Status	Attempts	Ref Sol	Naive Sol	Mutants Killed	Score
EASY	EX-3-EASY	VERIFIED	1	PASSED	FAILED (Good)	5/5	100.0%
MEDIUM	EX-4-MEDIUM	VERIFIED	1	PASSED	FAILED (Good)	5/5	100.0%

STRETCH	EX-7-STRETCH	VERIFIED	1	PASSED	FAILED (Good)	5/5	100.0%

EX-3-EASY (EASY) - Sandbox Execution Runs:

- **reference**: **PASS**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **naive**: **TEST_FAILURE** | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-1**: **TEST_FAILURE** | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-2**: **TEST_FAILURE** | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-3**: **TEST_FAILURE** | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-4**: **TEST_FAILURE** | Failing Test: **Test 3 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-5**: **TEST_FAILURE** | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`

EX-4-MEDIUM (MEDIUM) - Sandbox Execution Runs:

- **reference**: **PASS**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **naive**: **TEST_FAILURE** | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-1**: **TEST_FAILURE** | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-2**: **TEST_FAILURE** | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-3**: **TEST_FAILURE** | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-4**: **TEST_FAILURE** | Failing Test: **Test 2 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-5**: **TEST_FAILURE** | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`

EX-7-STRETCH (STRETCH) - Sandbox Execution Runs:

- **reference**: **PASS**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **naive**: **TEST_FAILURE** | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-1**: **TEST_FAILURE** | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-2**: **TEST_FAILURE** | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-3**: **TEST_FAILURE** | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-4**: **TEST_FAILURE** | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-5**: **TEST_FAILURE** | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning`

main.py

3. Template Compliance QA

Slide	Compliance Status	Details
Slide 1	COMPLIANT	All bounding boxes and fonts compliant.
Slide 2	COMPLIANT	All bounding boxes and fonts compliant.
Slide 3	COMPLIANT	All bounding boxes and fonts compliant.
Slide 4	COMPLIANT	All bounding boxes and fonts compliant.
Slide 5	COMPLIANT	All bounding boxes and fonts compliant.
Slide 6	COMPLIANT	All bounding boxes and fonts compliant.
Slide 7	VIOLATION	Code font size (14pt) below minimum allowed (18pt)
Slide 8	COMPLIANT	All bounding boxes and fonts compliant.
Slide 9	COMPLIANT	All bounding boxes and fonts compliant.